

Monologue (Topic: Body illusion effect)

Speaker:

大家好，今天我们要探讨一个非常有趣且富有启发性的心理学现象——身体错觉效应。你有没有想过，我们对自己身体的感知其实并不总是准确的？嗯，这就是身体错觉效应要告诉我们的。

Hello everyone, today we are going to explore a very interesting and insightful psychological phenomenon—the body illusion effect. Have you ever thought that our perception of our own body is not always accurate? Well, this is exactly what the body illusion effect highlights.

身体错觉效应是知觉错觉的一种表现，它揭示了大脑如何复杂而灵活地处理来自视觉、触觉等多种感官的信息，来构建我们对身体的认知。

The body illusion effect is a type of perceptual illusion, revealing how the brain processes information from multiple senses—such as vision and touch—in a complex and flexible way to construct our sense of the body.

简单来说，当不同感官的信息发生冲突时，我们的大脑就会产生错觉，甚至会让我们感觉一个外物成了自己身体的一部分。是不是很神奇？

Simply put, when information from different senses conflicts, our brain can create an illusion, even making us feel that an external object has become part of our own body. Isn't that fascinating?

说到这里，我想给大家介绍一个经典的实验——橡胶手错觉实验。这个实验是 1988 年由心理学家马修·博特文尼克和乔纳森·科恩设计的。

At this point, I'd like to introduce a classic experiment—the rubber hand illusion experiment. This experiment was designed in 1988 by psychologists Matthew Botvinick and Jonathan Cohen.

实验中，参与者的真实手被遮挡，视线中出现一只橡皮手。研究者用刷子同时轻触真实手和橡皮手。经过一段时间，参与者竟然会感觉橡皮手就是自己的一部分！

In the experiment, the participants' real hand was hidden from view, and a rubber hand was placed in sight. The researchers stroked both the real hand and the rubber

hand simultaneously with a brush. After some time, participants actually felt that the rubber hand was part of their own body!

更令人惊讶的是，当橡皮手受到敲击时，参与者会产生焦虑反应，好像真的受到了伤害一样。这个现象体现了大脑对身体边界的灵活调整。

Even more surprisingly, when the rubber hand was tapped, participants exhibited anxiety as if they were really being hurt. This phenomenon demonstrates the brain's flexible adjustment of body boundaries.

后来，研究者们将这一原理扩展到了全身错觉实验。参与者戴上头戴式显示器，从第一人称视角观察一个虚拟身体，同时接受同步的多感官刺激。

Later, researchers extended this principle to full-body illusion experiments. Participants wore head-mounted displays to observe a virtual body from a first-person perspective while receiving synchronous multi-sensory stimulation.

这种同步的视觉和触觉刺激让参与者产生了对虚拟身体的强烈拥有感，仿佛那个虚拟身体就是他们的真实身体。

This synchronous visual and tactile stimulation created a strong sense of ownership over the virtual body, as if it were their own real body.

那么，是什么因素促使这种错觉产生呢？首先，虚拟身体的解剖结构必须与真实身体相符。你能想象如果虚拟身体结构扭曲，我们还会产生拥有感吗？

So, what factors induce this illusion? First, the anatomy of the virtual body must match the real body. Can you imagine experiencing a sense of ownership if the virtual body were distorted?

其实不会。研究显示，结构正常的虚拟身体更容易让人产生身体拥有感。其次，感官刺激的时间和空间必须同步，否则错觉就难以形成。

Actually, no. Studies show that a properly structured virtual body more readily produces a sense of body ownership. Second, the timing and spatial alignment of sensory stimulation must be synchronized; otherwise, the illusion is difficult to form.

这些发现告诉我们，身体感知是大脑基于多感官信息动态整合的结果，而非固定不变的事实。

These findings indicate that body perception is the result of dynamic integration of multi-sensory information by the brain, rather than a fixed reality.

这对于设计领域有着深远的启示。传统的人机交互往往注重界面的美观和效率，但我们应更关注如何塑造用户的“身体图式”。

This has profound implications for design. Traditional human-computer interaction often focuses on interface aesthetics and efficiency, but we should pay more attention to shaping the user's "body schema."

比如，在虚拟现实，通过同步用户的视觉、触觉和动觉反馈，可以增强用户对虚拟化身的拥有感，从而提升沉浸感和任务参与度。

For example, in virtual reality, synchronizing visual, tactile, and proprioceptive feedback can enhance users' sense of ownership over avatars, thereby increasing immersion and task engagement.

在临床康复方面，身体错觉效应也有实际应用。例如，通过同步的视觉和触觉刺激，截肢患者可以对假肢产生所有权感，有助于缓解幻肢痛。

In clinical rehabilitation, the body illusion effect also has practical applications. For example, through synchronized visual and tactile stimulation, amputees can develop a sense of ownership over prosthetic limbs, which helps alleviate phantom limb pain.

这是一种非药物的有效干预方法，体现了心理学和技术结合的力量。

This is an effective non-pharmacological intervention, demonstrating the power of combining psychology and technology.

此外，基于身体错觉的设计还能引导用户的行为和态度改变。你是否想过，体验成为另一个身体，会对我们的偏见产生影响？

Moreover, designs based on body illusions can influence users' behavior and attitudes. Have you ever thought that experiencing being another body could affect our biases?

实验表明，当人们体验成为老年人的虚拟身体后，他们对老年人的隐性偏见显著降低。

Experiments show that when people experience being in a virtual elderly body, their implicit biases toward older adults are significantly reduced.

类似地，化身为珊瑚体验海洋酸化过程的用户，环保意识和亲环境行为意向也有了明显提升。

Similarly, users embodying a coral to experience ocean acidification show increased environmental awareness and pro-environmental behavioral intentions.

这说明身体错觉不仅仅是个心理现象，更是一个强大的工具，可以帮助我们理解自己，甚至促进社会积极变化。

This indicates that body illusions are not merely a psychological phenomenon, but a powerful tool to help us understand ourselves and even promote positive societal change.

嗯，大家有没有想过，我们未来的技术会如何利用身体错觉效应，打造更加自然和沉浸的交互体验呢？

So, have you ever thought about how future technologies could use the body illusion effect to create more natural and immersive interactive experiences?